Centralized Rineya Titan 0.6B: laptop execution revision

30 September 2026, America/Chicago. **The inherited 7.4 GiB admission guard was not measured Titan RAM demand. Exact GPU loading and frozen inference have now run successfully on this laptop, with observed process-tree RAM peaks below 2 GiB. The complete 512-token CE mechanics trial now passed all eight checks, four disposable updates, bitwise save/resume and unchanged-base verification. Its observed process-tree RAM peak was 1.7486 GiB. The earlier 1024 VRAM failure remains preserved. The observed RSS margin is only 1.39 MiB; sustained E1 fit, acquisition and retention remain unmeasured. No recipe is promoted.**

Exact primary student and source

Primary is Rineya-Titan-Experimental-1.0.0-devfreeze: 596,049,920 unique parameters, 310 serialized BF16 tensors, 28 layers, hidden size 1024, MLP 3072, 16 query heads, 8 KV heads, head dimension 128 and vocabulary 151936. Native loading verified the embedding tie and exact copied values. Declared context 40960 is not a qualified operating envelope. The static BF16 payload is 1,192,099,840 bytes, or 1.110 GiB.

Weight SHA256: `537201676814ade103c4efa63f5ebe758ea309971803c4ad633dbca5b6b531b7`. Config SHA256: `660db3b73d788119c04535e48cf9be5f55bc3100841a718637ae695b442f27dd`. Tokenizer SHA256: `aeb13307a71acd8fe81861d94ad54ab689df773318809eed3cbe794b4492dae4`. Template SHA256: `a55ee1b1660128b7098723e0abcd92caa0788061051c62d51cbe87d9cf1974d8`.

The exact local source remains the published Titan devfreeze checkpoint in Chat-Bubble/Rineya_Published_Cores. The dedicated execution branch is research/post-training-mimo-centralized-v1, descended from verified parent `03db218ceebc61a2bc04f2482cf7e094045e01c1`. No parent/main merge or history rewrite occurred. Historical 344M evidence remains preserved and outside primary training. The first two native runs used `79ad1a61ba1bb16b93ccc9f0cd27c0e22b9bb00c`. The tested streamed implementation was committed at `4526d705` (abbreviated; full identity is retained in its run receipt and Git). The measured unpadded trial used full producer `a4454e17009c9d3aadd87e2b9b633c2969f5be00`. The 512 successor native source is `d5640306dac3ea33b862e77bb1eb932a980a9e4f`, with evidence checker at `431a9deb`. The complete CE512 trial used producer `15901919825dc1855020212edcaf6830ee7caef1`; its actual evidence checker passed. Final documentation and launcher identities are provided by the delivery commit.

Why the RAM policy changed

The user explicitly authorized changing the RAM requirement and plan for a 16 GB laptop, with priority given to VRAM-heavy work. Source audit traced the old 7.4/6 GiB available-memory floors to PRLC project defaults for a GPU model plus frozen CPU reference. The central loader has no full CPU reference: it constructs BF16 weights directly on CUDA and hashes through small chunks. Earlier admission stops therefore established compliance with a policy, not a measured requirement for 7.4 GiB free RAM or a finding that Titan cannot fit.

Rank-8 q/v adapters contain 1,146,880 FP32 parameters: 4.375 MiB. Parameters, gradients and two Adam moments total approximately 17.5 MiB, primarily on GPU. Activation and framework peaks need measurement for each operating envelope. Quantization or CPU optimizer offload has not been added to this first route.

AMENDMENT_009 registered a bounded pilot with 2 GiB system reserve and 2.5 GiB worker RSS cap. Admission reserves all remaining permitted worker growth rather than a fixed 7.4 GiB buffer. After actual loading succeeded, AMENDMENT_010 separated backward/resume memory measurements from the unresolved padded-forward/sampler qualification. The numerical tolerances remain frozen.

AMENDMENT_011 replaces mapped CPU tensor access with a fixed 2 MiB read buffer. It validates the BF16 header, offsets, shapes and complete payload coverage, copies synchronously to GPU and checks every loaded tensor hash. Its named profile keeps 1.25 GiB available system RAM and caps process-tree RSS at 1.75 GiB. Admission is 1.25 + max(0, 1.75 - current RSS) GiB: a 3 GiB before-worker budget. These are prospective trial bounds. A later admitted trial copied all

310 exact GPU tensors and completed 256/512 backward and optimizer fixtures within the RAM cap. The complete 1024 scope failed its VRAM floor. A separately registered CE512 trial subsequently passed complete save/resume and frozen-base verification. External 100 ms monitoring, Windows containment, GPU 4/0.5 GiB floors and disk 4 GiB floor remain active. No user application was terminated.

Native results and remaining limits

Attempt	Measured outcome	Peak process RAM	Interpretation
Load pilot	Exact load and finite 256-token forward; 10.406 s worker /11.219 s supervisor	1.928 GiB	Load/inference scope only; CUDA allocation peak 1.400 GiB
Full laptop v1	Exact load and 32 zero-adapter cases, max difference 0; padding check failed at 15.656 s /16.546 s	1.951 GiB	All recorded resource checks passed; no backward/update/generation
Instrumented full v2	Preload RAM 2.515 GiB versus dynamic 3.831 GiB; 7.891 s /8.719 s	0.673 GiB	No model work; numerical error magnitude not remeasured
Streamed load v1	Preload RAM 2.110 GiB versus dynamic 2.331 GiB; 5.078 s /5.875 s	0.673 GiB	New loader not reached; no model work
Streamed unpadded v1	Exact load, numerical gates and 256/512 updates; 32.000 s /33.172 s	1.740 GiB	1024 backward returned; free VRAM 0.161 GiB below 0.5 GiB reserve
Complete CE512 v2	Eight checks, four updates, bitwise resume and unchanged base; 32.297 s /33.360 s	1.7486 GiB	Single unpadded CE through 512; sustained E1 remains unmeasured

The full attempt failed its first left-padding likelihood comparison against maximum error 0.03125 and mean error 0.01. Its original receipt did not store the error magnitude. Failure-only telemetry now preserves errors and uses null for nonfinite entries; the instrumented retry stopped before loading. Source inspection found aligned positions and plausible masks. BF16 shape-dependent rounding is a hypothesis, not an established cause. No threshold was relaxed.

Two non-author assistant reviews checked native source/config/journal/cost and scope consistency; this is not human approval. Native verified loads total four and disposable optimizer calls total six. Completed-step input positions total 2560; actual training forwards used 3584 positions across the two latest trials. No generation, teacher calls or capability measurements occurred. The historical 71 stdlib controls and nine actual tiny CPU loader controls passed; earlier 12 tiny numerical controls remain a separate run. One historical-helper test initially failed after adding a new dependency; the corrected fixture verifies actual old dependencies and rejects inventing a dependency in an old commit. The failed log/receipt remains preserved. CPU controls do not establish GPU fit.

App-independent execution and detailed sequence

The corrected standalone controller completed its admission window after 601.063 seconds. It recorded 256 observations, with 0.581-0.955 GiB available, and never invoked a native supervisor. Its process is terminal and the raw wait receipts are preserved. A later OS reading was 1.397 GiB with no Python workers. Headroom subsequently rose to 5.221 GiB; the controller measured 5.118/5.121 GiB and admitted the streamed unpadded trial. No user application was terminated. A regular PowerShell launcher can run outside the desktop app. It waits at most ten minutes for two observations at least 3.0625 GiB, including supervisor allowance, then invokes one contained 30-minute scoped qualification trial. The latest 512 successor admission window was deliberately capped at 60 seconds and returned after 62.552 seconds, with 27 observations of 1.441-1.853 GiB. No native supervisor was invoked; the controller is verified terminal and its wait receipts are preserved. A later actual headroom change to 4.303 GiB enabled two admission readings of 4.275/4.278 GiB and the complete CE512 trial. That controller is terminal after 37.248 seconds. It aborts on source/config changes, resource failure, timeout or numerical failure and records durable results. Desktop working-set observations near 2 GiB do not guarantee that closing it reclaims exactly that amount. Actual headroom is checked before model work.

The earlier backward_resume resource stage retains the exact loader, base and zero-adapter comparisons, positional EOS/PAD label mechanics, 256/512/1024 gradients, five disposable optimizer calls, checkpoint save/reload and bitwise within-stage resume. It omits unresolved padded-forward/causality and sampler checks and explicitly records the changed sampler RNG trajectory. Its distinct resource-only PASS cannot satisfy full E0 or authorize E1. See LAPTOP_RUNBOOK and the launcher files.

After resource and numerical qualification, bind measured laptop successors for native collection, queue and E1. Prioritize one resident GPU worker, pretokenized streamed microbatches, GPU adapters/optimizer state, adapter-only checkpoints and sequential evaluation panels. Avoid retaining a second CPU model or concurrent CPU-heavy data preparation. Register activation checkpointing or shorter contexts only if measured VRAM requires them. Existing E1 plans remain historical until rebound to qualified current source/runtime.

Scope matching the actual SFT inputs

AMENDMENT_012 registers UNPADDED_SINGLE_SEQUENCE_CE_V1. Prepared E1 data require microbatch one, zero padding, all-one masks and completion-only labels. The new scope validates those boundaries, omits a redundant validated all-one mask for primary forwards and rejects cache, custom positions and packing metadata. Semantic data eligibility still requires actual row/token/label binding; IDs alone cannot prove a sequence is eligible.

The scoped worker adds mandatory native 80-token unpadded causality (bitwise 77-token prefix agreement after perturbing the last three tokens) and an actual explicit-all-one-mask comparison at the unchanged 0.03125/0.01 likelihood bounds. It retains EOS labels, causal CE, finite confined gradients, all three context profiles, five disposable calls, resume and unchanged-base checks. Padding, batched/cached inference, sampled-policy likelihood and evaluation capture remain unsupported or unqualified. The known full E0 padding failure remains preserved.

The actual trial passed bitwise 77-token causality and all-one-mask likelihood with maximum and mean error zero. Finite confined gradients, causal CE and updates completed at 256/512. The 512 single-step profile had a 3.089 GiB CUDA allocation peak and 0.203-second duration; this is a mechanics fixture, not sustained training throughput. At 1024, backward and gradient checks returned, then the VRAM guard stopped before the third optimizer call. Peak allocated/reserved VRAM were 6.254/6.639 GiB; minimum sampled host headroom was 3.032 GiB. All 387 accounted operations returned; no checkpoint save/reload or final frozen-base check occurred.

AMENDMENT_013 registers UNPADDED_SINGLE_SEQUENCE_CE512_V1 with contexts 256/512, four disposable updates and bitwise 512-token resume. Its hard input ceiling is 512 tokens. Prepared TRAIN sequences are at most 398 tokens, so the complete data dose and targets fit without truncation. Model math, eval mode, optimizer, RNG preservation and numerical/resource floors remain unchanged. The narrower fixture changes the declared workload; resume is checked within its own stage, without asserting cross-stage RNG equivalence. Activation checkpointing remains a separately registered future option for longer contexts. Long-context evaluation and preservation still require independent native qualification.

The scoped consistency checker requires explicit selection of the new 512 stage, with distinct config SHA and PASS status. It binds 393 calls, 26608 positions, 1792 training positions, 75 resource samples and 865 journal records; timing measurements must cover the recorded intent/return intervals. It verifies native checks, current/committed source, runtime, inventories, containment, opaque resume bytes and exact source assets. It never imports the model framework or grants E1. This is local byte identity and consistency, not authenticated execution proof.

All 107 current stdlib controls passed in 6.107 seconds (process 6.538 seconds). Separate non-author assistant reviews found no blocking native/checker source issue; this is not human or native approval. Old/failed/resource-only statuses and synthetic values cannot qualify a bundle. Windows PowerShell AuditOnly also passed. The default launcher is now unpadded_ce512. The subsequent complete native CE512 result and its 37 byte/source bindings passed the consistency checker in 2.207 seconds. Independent actual metadata/source review found no blocking contradiction. All 393 calls returned: 71 forwards, four backwards/updates and one save/reload; 26608 positions, 1792 training positions, 75 resource samples and 865 journal records. The eight native checks all passed.

Observed process-tree RSS was 1.748646 GiB, only 1.386719 MiB below the 1.75 GiB cap. Minimum sampled host headroom and GPU free memory were 2.498425/3.326172 GiB. Peak allocated/reserved CUDA memory were

3.098094/3.474609 GiB. Timers agree with the ordered journal. This short fixture does not measure gradient accumulation, data-reader overhead, sustained training, response capture or preservation. No controller or model worker is running. E1 remains unauthorized.

Actual TRAIN compatibility with the low-RAM interface

The existing numeric binder only syntax-checked source-row hashes. A prospectively frozen 28-file contract now joins every actual TRAIN source body to its token/label array and the CE512 consumer interface. Pure deterministic helpers are byte-pinned before import, rebuild semantic parents/TRAIN allocations and independently verify prompts and targets. Language joins complete curated conversations, original-body lineage and measured token profiles; all 81 prior assistant turns remain masked. One conversation can contain multiple turns without being sequence packing.

The actual committed producer `e717d87a59263e84db7b555dd902f48a7b928eac` passed all 5120 rows, 680532 input and 63546 supervised positions, maximum length 398. No row or target was truncated or rewritten. The unchanged historical data SHA is `51b87fe4fd70d2fb23d16f270d1023716172dbbe1c88efd79dff8c3548623d6d`; source/array join SHA is `555776a347ea7a82e7a111b90ab041f5947bb0e54bd32c8b306e9064fbd5fc68`. The production audit took 3.329 seconds (process 3.801 seconds), with a 39.004 MiB reader working-set peak. Eighteen new stdlib controls passed in 0.270 seconds; the earlier 107-control run remains separate. No model/tokenizer/weight access, new data generation, GPU work or closed-bank access occurred in this audit.

Independent review found and fixed a production provenance gap: receipt creation now snapshots committed producer/manifest bytes upfront and rechecks them, HEAD and the budget before publication. Dirty, untracked and changed producers are rejected. Resource checks are cooperative with OS high-water memory observations; there is no hard CPU containment. Language original TRAIN origin and prior quality selection rely on frozen earlier provenance; this is not new human grading, retokenization or complete pretraining exclusion. The data pass leaves every E1 and native launch grant false. Actual capture/decoding, preservation, language calibration, long-context runtime and current E1 execution remain prerequisites.

Prepared science and milestone audit

Preparation remains available: 4096 acquisition TRAIN parents, 768 deterministic and 256 curated language replay rows; 5120 total sequences with 680532 nonpadding and 63546 supervised tokens. The historical seed-101 plan has 256 steps, each 16 acquisition plus 4 replay, 13 warmup steps, final LR zero and final-only selection. Later seeds 211/307 require its registered DEV screen.

The DEV prompt/scorer binding contains 832 rows over 544 parent groups and 890874 prompt tokens, with evaluator references separate. Language has 64 DEV/16 CAL candidate worlds and unassigned review packets; grading remains unqualified. Long-context DEV has 32 worlds and 256 related rows at exact 2024/4072 prompt lengths; native 2048/4096 preservation remains unqualified. Parent grouping, source-quality exclusions, original failures and scoped fingerprint checks are retained in their existing receipts and ledgers. CHECK/FINAL remain closed.

Milestone item	Current evidence	State	Remaining work
Current state	Complete CE512 mechanics and actual TRAIN joins	Partial progress	Preservation/capture/current E1
Exact model	Titan devfreeze; 596049920 parameters	Bound	Same checkpoint retained
Git/source	Complete CE512 producer 15901919	Committed	Delivery commit records final docs
Verified	Eight CE512 checks; 5120 source/array/consumer joins	Scoped	Full E0 and E1 not passed
Unmeasured	Sustained E1 fit, preservation and capability gain	Open	Current capture and E1 successors

Milestone item	Current evidence	State	Remaining work
Completed experiment	Complete native CE512 mechanics	Recorded	Prior failures and partial work retained
Exact result	Latest RAM peak 1.7486 GiB; minimum GPU free 3.326 GiB	Measured	Sustained worker overhead
Acquisition effect	Disposable mechanics only	Unmeasured	Qualified E1
Retention effect	No scored native preservation	Unmeasured	Capture and grading
Compute/memory	Six workers 103.328 s; supervisors 108.891 s	Measured wall	No sustained throughput inference
Failure modes	Padding, 1024 VRAM reserve and RAM admission	Recorded	No tolerance or guard auto-relaxation
Interpretation	7.4 GiB was inherited reserve, not model demand	Source audit	Complete training envelope still needs proof
Ruled out	Admission failures alone prove Titan needs 7.4 GiB free	Invalid inference	No efficacy conclusion
Next experiment	Qualified capture and graded preservation	Open	Current native plans and E1 fit
Heavy compute justified	Bounded qualification	Conditional	E1 waits for scientific gates

The original capability/retention research objective remains incomplete. Current best remains unchanged Titan with null acquisition and retention effects. No preparation or partial resource pass promotes a recipe. Historical blocked audits and 7.4/6 GiB configs remain evidence of earlier conditions; they are superseded operationally by the user's authorized laptop route, not rewritten into successes.